

Regular Graph (Open Chain)



A — B

A — C

A — D

B — C

B — D

Length 3

C — D

Length 2

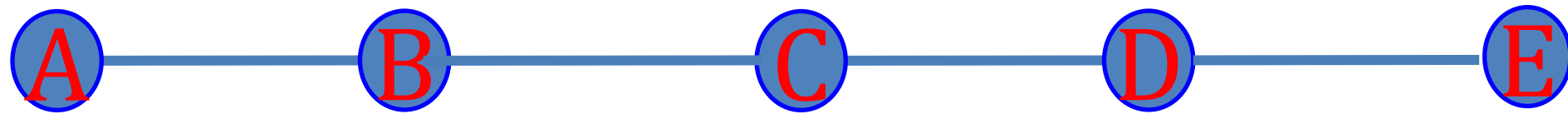
$4-3=1$

Length 1

$4-2=2$

$4-1=3$

Regular Graph (Open Chain)



A — B

A — C

A — D

A — E

Length 4

B — C

B — D

B — E

5-4=1

Length 3

C — D

C — E

5-3=2

Length 2

D — E

5-2=3

Length 1

5-1=4

$$\frac{2}{(n(n-1))} \sum_{j=1}^{n-1} j(n-j) =$$

$$\frac{2}{n(n-1)} (1 \times (n-1) + 2 \times (n-2) + 3 \times (n-3) + \cdots + (n-1) \times (n-n+1))$$

$$= \frac{2}{(n(n-1))} (n(1+2+3+\cdots+n-1) - (1^2+2^2+\cdots+(n-1)^2)) = \frac{2}{(n(n-1))} \left(\frac{n(n-1)n}{2} - \frac{(n-1)n(2n-2+1)}{6} \right)$$

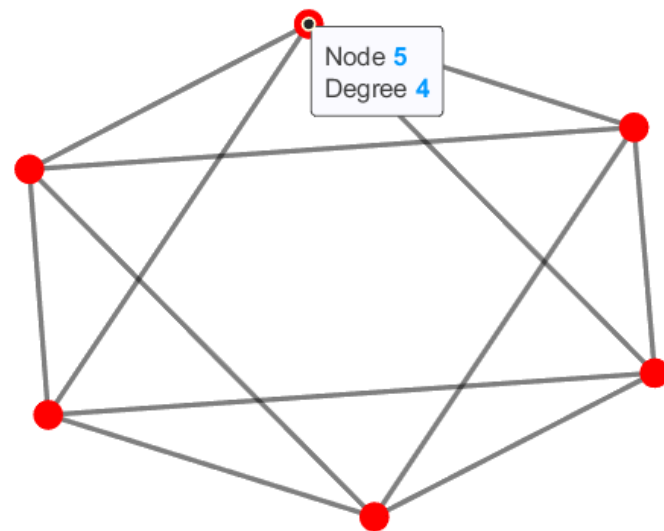
$$= \left(n - \frac{(2n-1)}{3} \right) = \frac{n+1}{3}$$

Star Graph

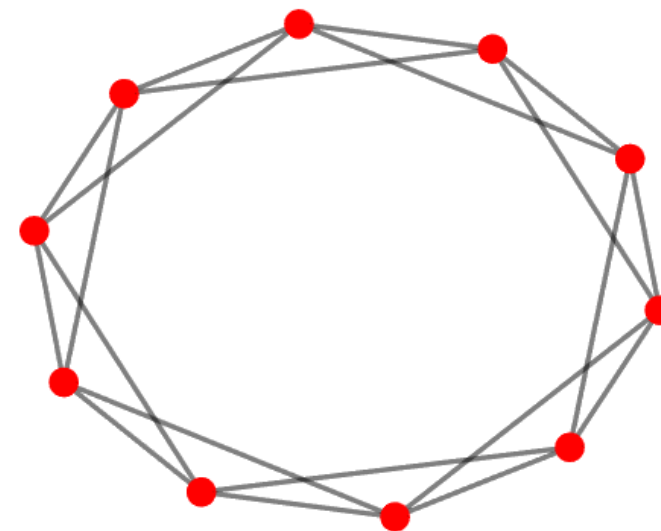
Q: Find the average pathlength, clustering coefficients of star graph.
Change network size , derive mathematically and explain.

Regular Graph (Ring)

$N=6$



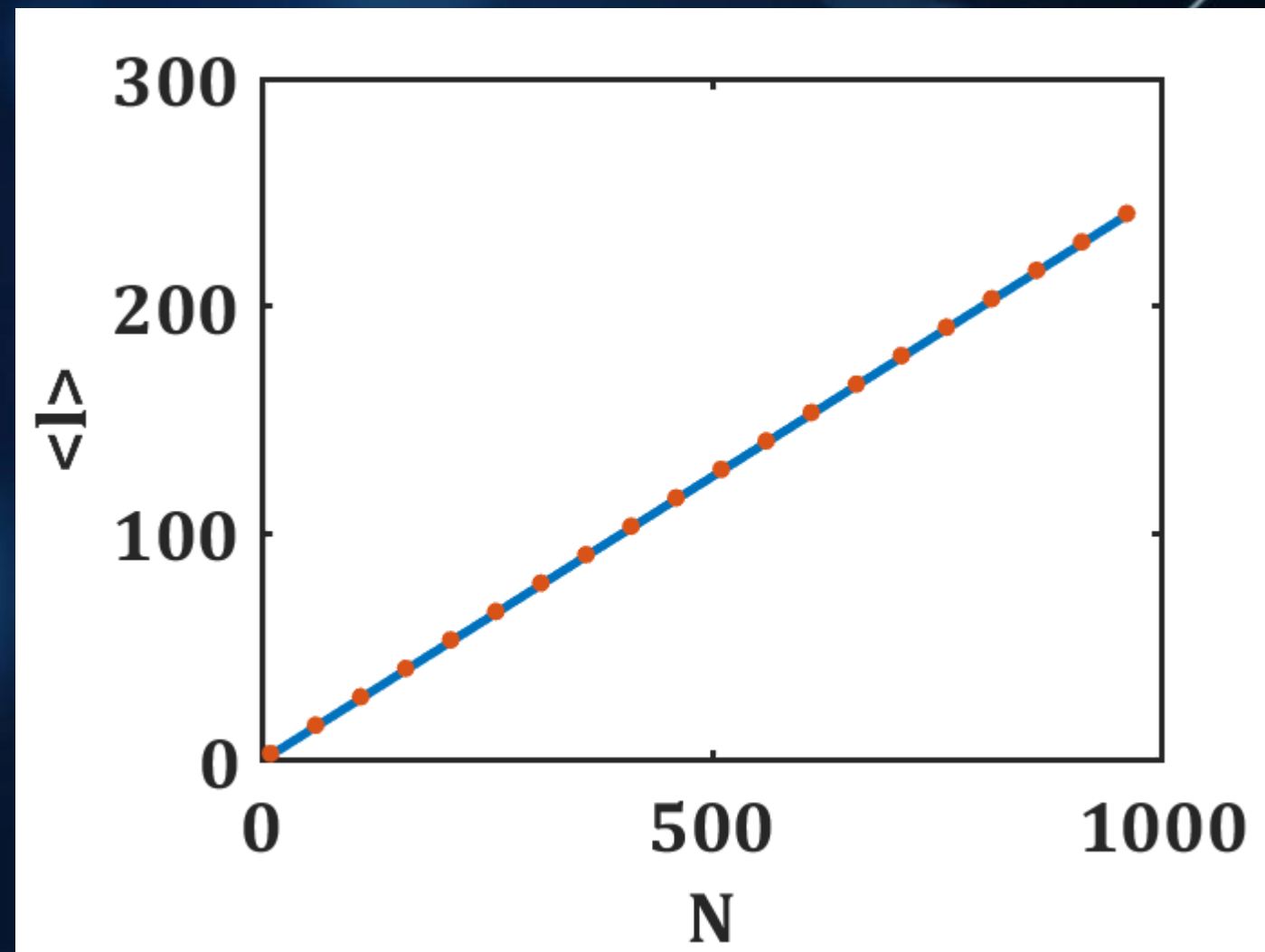
$N=10$



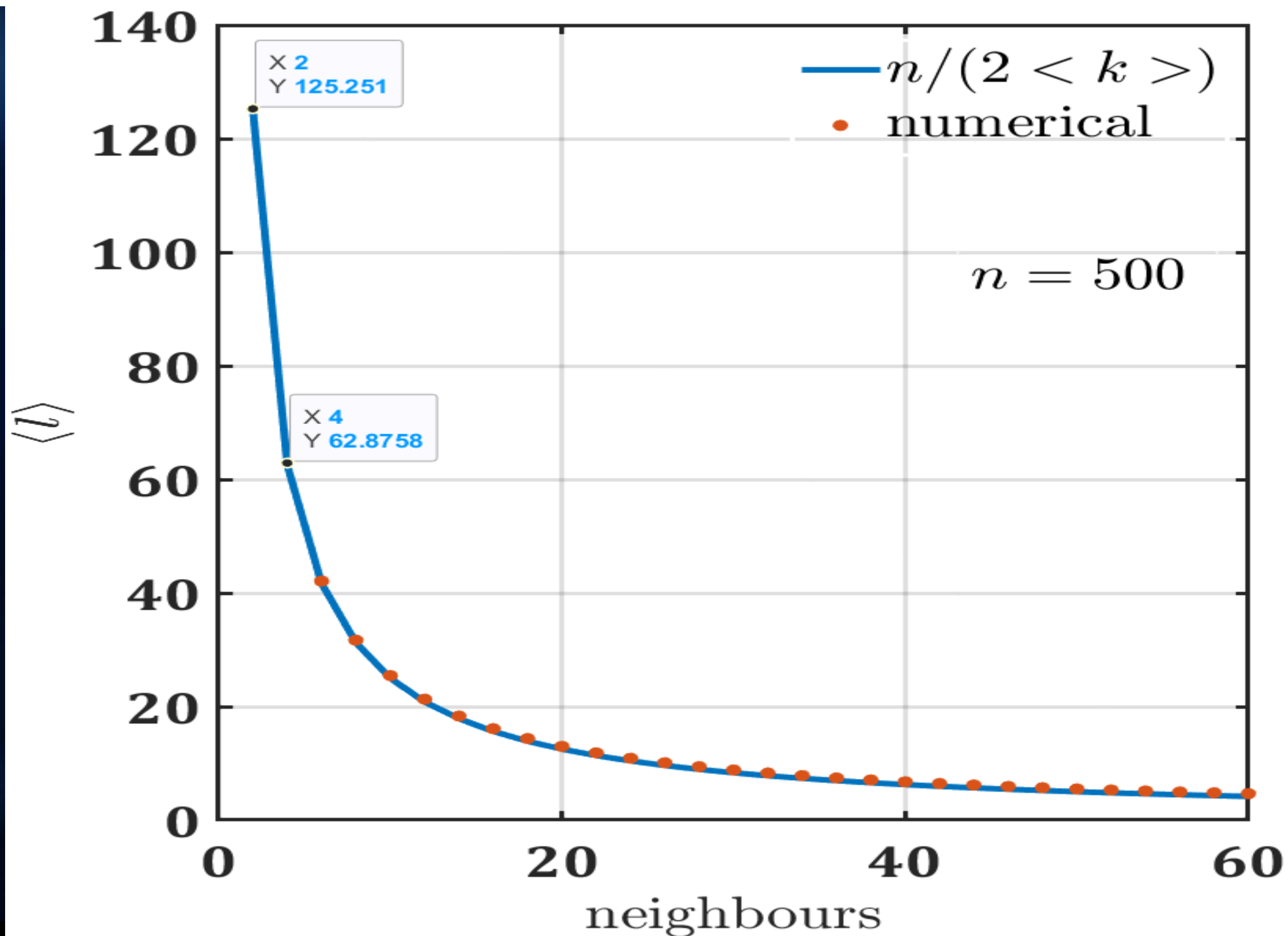
Q: What will be the average pathlength?

Regular Graph (Ring)

Q: What will be the average pathlength?



Regular Graph (Ring)



Regular Graph (Ring)

```
n=100;  
ZZ=[];  
for n=[6 10]  
    n  
    c=2;  
    %for c=1:3:20  
    A = smallw(n,c,0);  
    A=zeros(n,n)+A;  
    %A=A+A;  
    deg=sum(A);  
    avg_deg=mean(deg);  
    l=pathlength(A);  
  
    G=graph(A);  
  
    figure; plot(G);  
    deg = degree(G);  
end
```

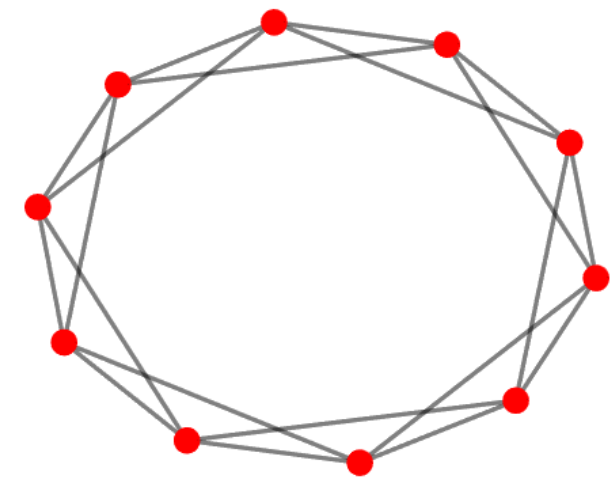


ONE DIMENSIONAL LATTICE: nodes on a ring

$$P(k) = \delta(k - 4) \quad k = 4 \text{ for each node}$$

$$C = \frac{1}{2} \text{ for each node if } N > 6$$

$N = 10$



ONE DIMENSIONAL LATTICE: nodes on a ring

$$P(k) = \delta(k - 4) \quad k = 4 \text{ for each node}$$

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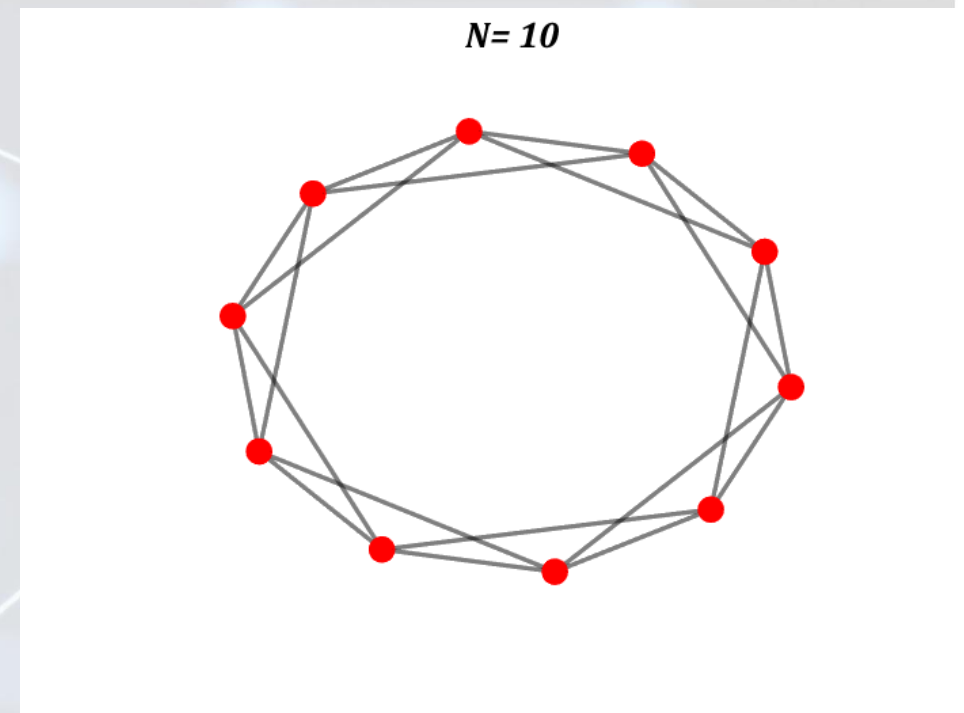
$$1 + \sum_{l=1}^{l_{\max}} 4 \approx N$$

$$\text{Thus, } l_{\max} \approx \frac{N}{4}$$

$$\langle l \rangle = \frac{4 \sum_{l=1}^{l_{\max}} l}{N} = \frac{4 \left(1 + 2 + \dots + \frac{N}{4} \right)}{N} \approx \frac{N}{2 * 4}$$

The average path-length varies as
Constant degree, constant clustering coefficient.

$$\langle l \rangle \approx N; N \gg 1$$



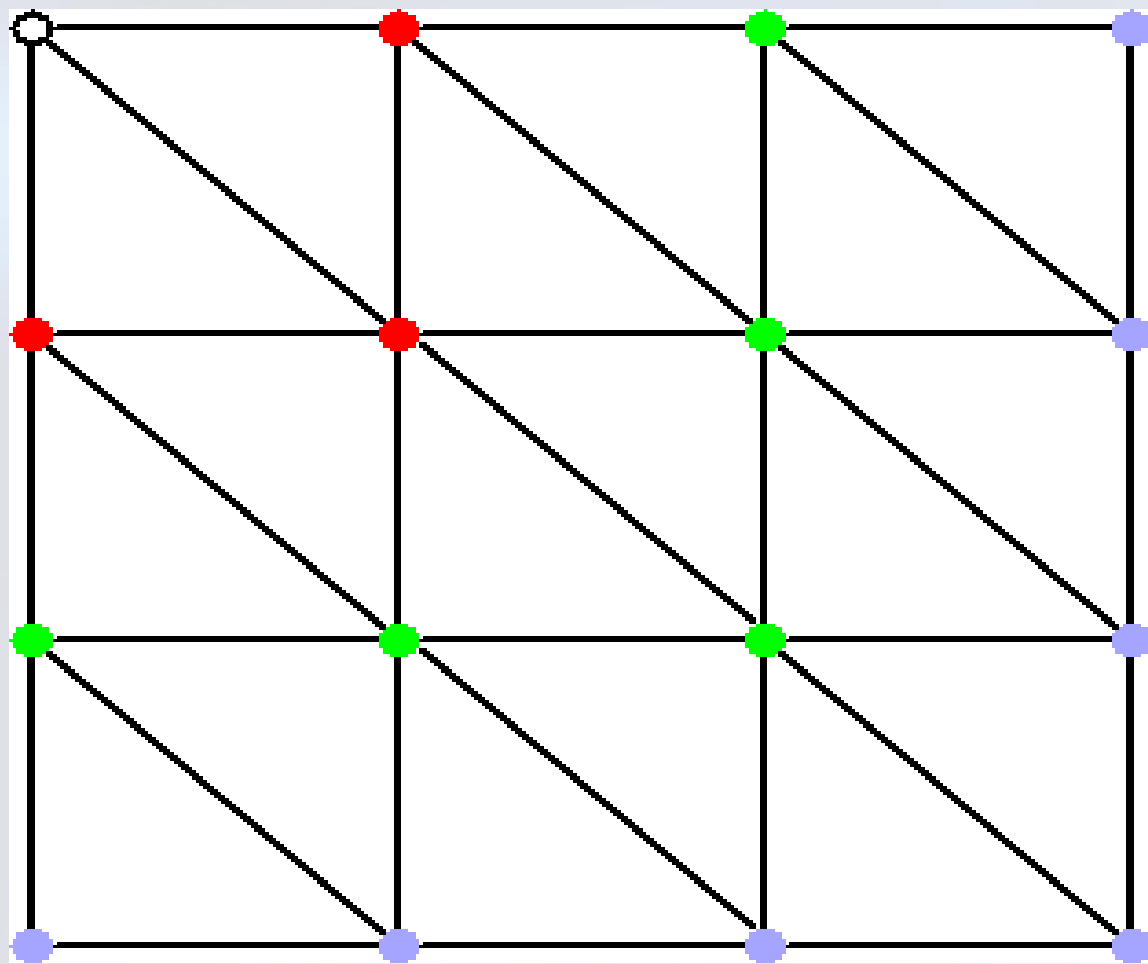
TWO DIMENSIONAL LATTICE

Example 3 *2D square lattice graph.* We can label vertices $v_{i,j}$ where i and j are integers (consider i, j are between 1 and n , so there are $p = n^2$ vertices). The distance between two vertices is $d(v_{i,j}, v_{k,\ell}) = |i - k| + |j - \ell|$. There are $\frac{n^2!}{2(n^2-2)!} = \frac{n^2(n^2-1)}{2}$ pairs of vertices. Thus the average path length is

$$\begin{aligned} \frac{1}{n^2(n^2-1)} \sum_{i,j,k,\ell=1}^n (|i-k| + |j-\ell|) &= \frac{1}{n^2(n^2-1)} \left(2n^2 \sum_{i,k=1}^n |i-k| \right) \\ &= \frac{2n^2}{n^2(n^2-1)} \left(\frac{1}{3} (n+1)n(n-1) \right) \\ &= \frac{1}{3}n = \frac{1}{3}p^{1/2}. \end{aligned}$$

Other than at the edges, each vertex is adjacent to 4 vertices, none of which are adjacent to each other, so it has clustering 0.

TWO DIMENSIONAL LATTICE



$$P(k) = \delta(k - 6) \quad k = 6 \text{ for inside nodes}$$

$$C = \frac{6}{15} \text{ for inside nodes}$$

$$1 + \sum_{l=1}^{l_{\max}} 6l \approx N \Rightarrow l_{\max} \propto N^{0.5}$$

$$\langle l \rangle \approx L \approx N^{1/2}$$

$$\langle l \rangle \approx N^{1/D}$$

In general, the average distance varies as $\langle l \rangle \approx N^{1/D}$ where D is the dimensionality of the lattice. Constant degree (coordination number), constant clustering coefficient.

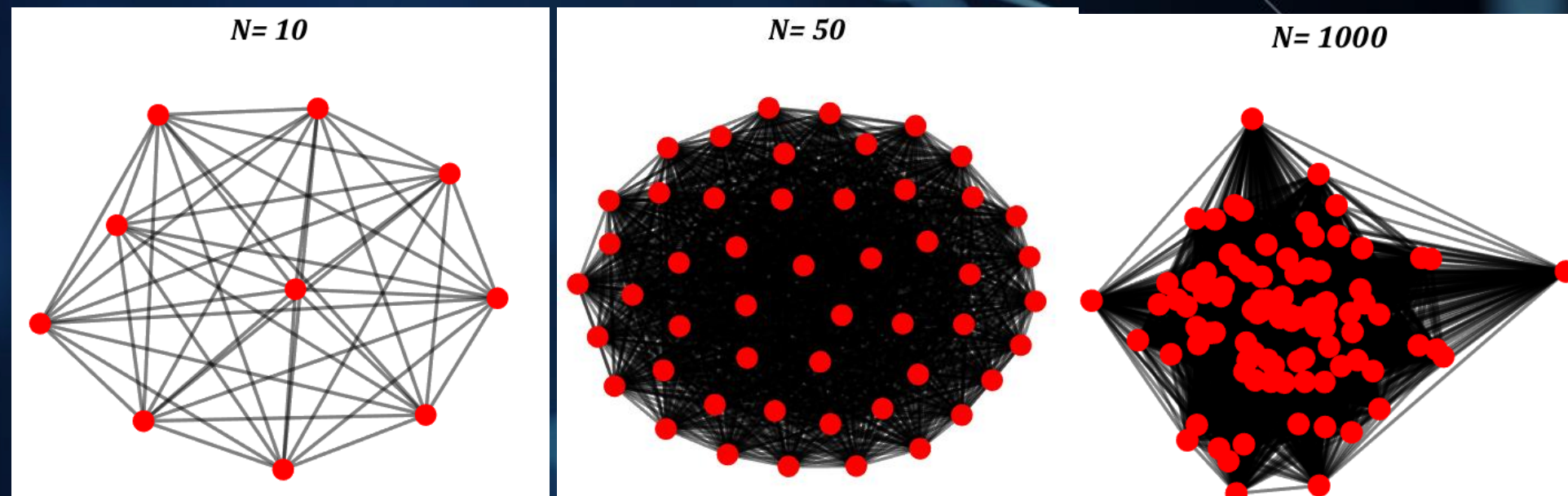
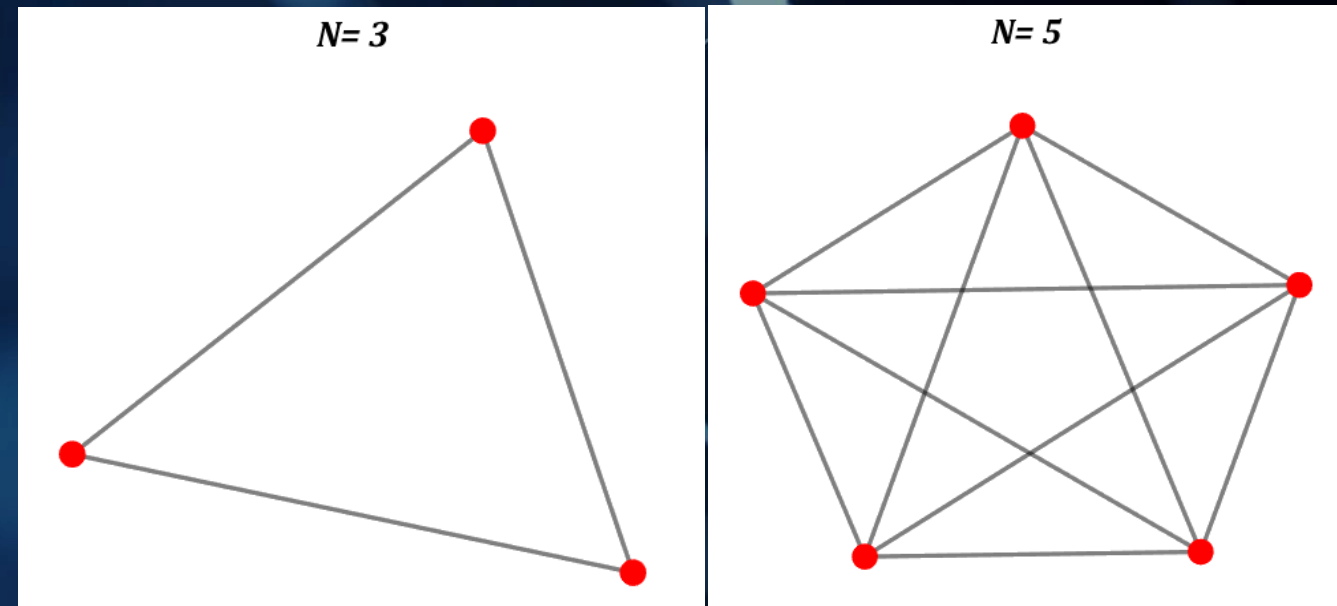
Complete Graphs

Complete Graph

- Connected
- Each node is connected to all other nodes

Complete Graphs

Complete Graph



Complete Graphs plot

Complete Graph

```
for N= [3 5 10 50 1000]
A=ones(N,N);
A(1:N+1:N*N)=0;
G=graph(A);
figure; plot(G);
deg = degree(G);
%deg = degree(h2);
nSizes = 8+4*sqrt(deg-min(deg)+0.5);
% NC=cell2mat(data(j,7));
nColors = rand(N,1)*0+3;%NC(:,3);% deg;
%%
%p=plot(G,'MarkerSize',nSizes,'NodeCData',...
% nColors,'EdgeAlpha',0.5);
p=plot(G,'MarkerSize',nSizes,'EdgeAlpha',0.5);
p.NodeFontSize=0.3;
p.NodeFontWeight='b';
p.EdgeColor='k';
```

Complete Graphs plot

Complete Graph

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% nColors,'EdgeAlpha',0.5);
p=plot(G,'MarkerSize',nSizes,'EdgeAlpha',0.5);
p.NodeFontSize=0.3;
p.NodeFontWeight='b';
p.EdgeColor='k';

p.LineWidth=2;
p.NodeColor='r';
A=[];
set(findall(gcf,'-property','FontSize'),'FontName','Cambria',...
'FontSize',24,'linewidth',2.0,'fontweight','b')
set(gcf,'InvertHardCopy','off','Color','white');

pp=mat2str(N);
[t]= title(sprintf('N= %s', pp));

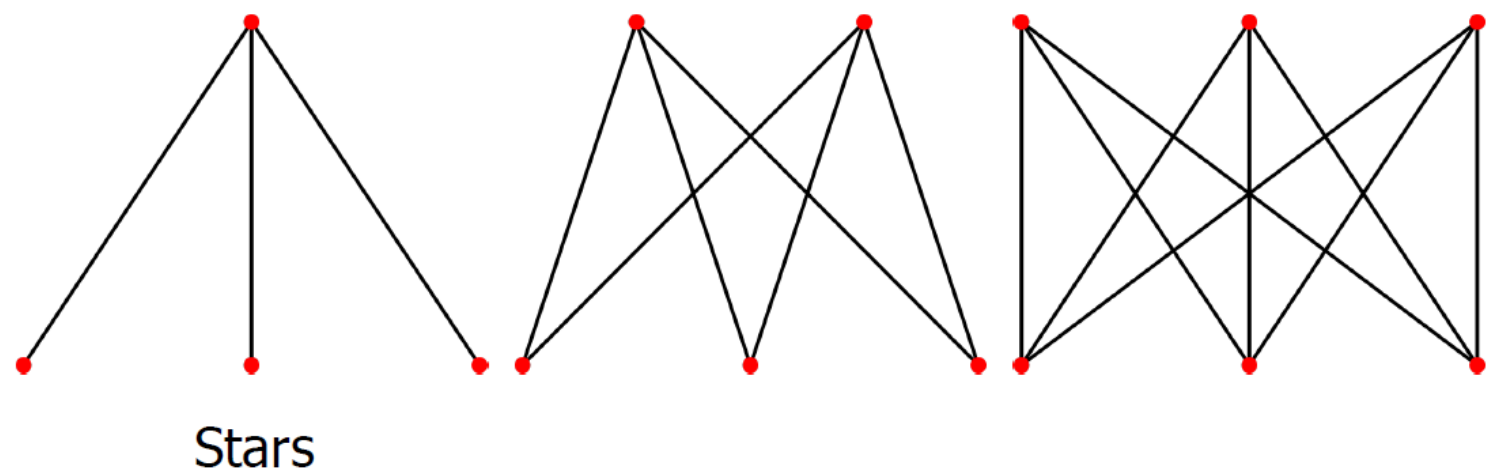
% title(['Time= ',num2str(pp), 'Sec'])
% pp=[];
t.FontSize = 16;
t.FontAngle = 'italic';
axis off
end
```

Complete bipartite Graphs

Complete bipartite Graph

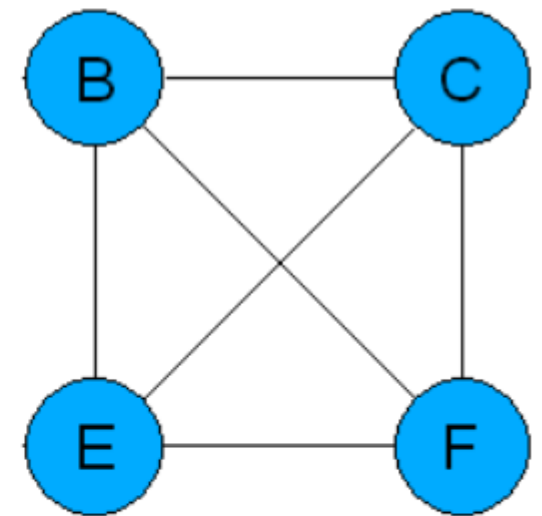
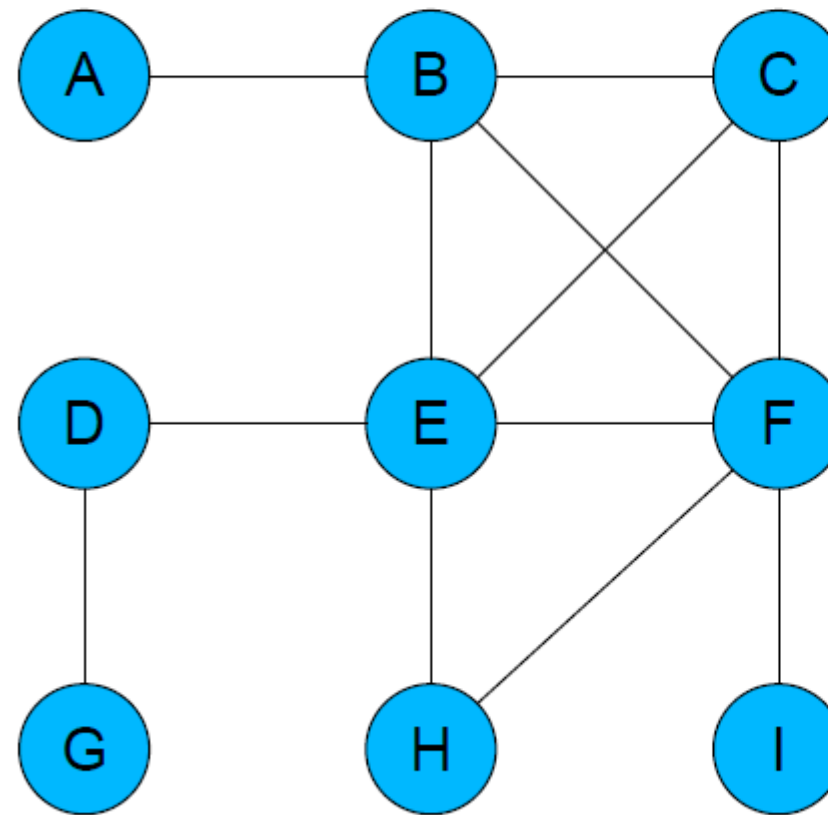
Complete Bipartite Graph

- Bipartite Variation of Complete Graph
- Every node of one set is connected to every other node on the other set



Cliques

A **clique** is a maximum complete connected subgraph.



Spanning Tree

Spanning tree

- Let G be a connected graph. Then a ***spanning tree*** in G is a subgraph of G that includes every node and is also a tree.

